Day 1 – Cybersecurity Fundamentals

What is Cybersecurity?

Cybersecurity is the practice of protecting computers, mobile devices, servers, networks, applications, and data from cyber attacks, theft, or unauthorized access.

Uses of Cybersecurity

- Protect personal data such as photos, passwords, and bank details
- Secure company systems and infrastructure
- Protect websites and applications
- Defend networks from hackers and malware

Common Cybersecurity Threats

1. Phishing Emails

Phishing is a cyber attack where attackers send fake emails or messages to trick users into revealing sensitive information.

Goals of Phishing

- Steal passwords
- Collect bank details
- Spread malware
- Trick users into clicking malicious links

Common Targets Used by Attackers

- Banks
- Delivery companies
- Social media platforms
- Employers

Prevention

- Verify sender email addresses
- Avoid clicking suspicious links
- Enable Two-Factor Authentication (2FA)
- Never share passwords through email

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2. Viruses and Ransomware

Virus

A virus is malicious software that spreads between files or systems and can:

- Damage files
- Slow down computers
- Steal information

Ransomware

Ransomware encrypts files and demands payment to restore access.

Symptoms of Ransomware

- Files become unreadable
- Ransom message appears on screen

Common Infection Methods

- Fake downloads
- Cracked or pirated software
- Malicious email attachments
- Unsafe USB devices

Protection Methods

- Keep regular backups
- Update operating systems and software
- Use antivirus software
- Avoid pirated software

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3. Password Attacks

Password attacks are attempts to steal or guess user passwords.

Common Methods

Brute Force Attack

Trying thousands or millions of passwords automatically.

Dictionary Attack

Using common passwords such as:

- 123456
- password
- qwerty

Protection Methods

- Use strong passwords
- Enable Multi-Factor Authentication (MFA)
- Avoid reusing passwords
- Use password managers

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4. Data Leaks

A data leak occurs when confidential or private information becomes publicly exposed or stolen.

Examples of Leaked Data

- Email addresses
- Passwords
- Phone numbers
- Bank details

Causes

- Weak security
- Human error
- Hacking attacks
- Cloud misconfiguration

Prevention

- Encrypt sensitive data
- Restrict access permissions
- Monitor systems regularly

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5. Website Hacking

Website hacking involves attacking websites to:

- Steal data
- Deface web pages

- Spread malware
- Gain control of servers

Common Website Attacks

SQL Injection

Attackers insert malicious SQL commands into forms or URLs.

Cross-Site Scripting (XSS)

Attackers inject malicious JavaScript into web pages.

Admin Panel Attacks

Attackers attempt to access admin panels using weak passwords.

File Upload Exploits

Malicious files are uploaded while pretending to be images or documents.

Protection Methods

- Validate user input
- Use strong passwords
- Update software regularly
- Use firewalls and security monitoring

Security Operations Center (SOC)

What is SOC?

A Security Operations Center (SOC) is a centralized cybersecurity team responsible for monitoring, detecting, analyzing, and responding to cyber threats in real time.

Main Goal of SOC

The primary goal of a SOC is:

> Detect cyber attacks early and stop them before major damage occurs.

Systems Protected by SOC Teams

- Computers
- Servers

- Employee accounts
- Cloud systems
- Websites
- Networks

How a SOC Works

Step 1 – Log Collection

Every device generates logs such as:

- Login attempts
- File access events
- Failed password attempts
- Network connections
- Software activity

SOC teams collect these logs for monitoring and analysis.

Step 2 – SIEM Analysis

SOC teams use SIEM platforms.

SIEM Meaning

SIEM = Security Information and Event Management

Popular SIEM Tools

- Splunk
- IBM QRadar
- Microsoft Sentinel
- Elastic Security

Functions of SIEM

- Collect logs
- Correlate events
- Detect suspicious activities
- Generate alerts

EDR and XDR

What is EDR?

EDR stands for Endpoint Detection and Response.

It monitors and protects endpoints such as:

- Laptops
- Desktops
- Servers

Functions of EDR

- Detect malware
- Monitor system activity
- Analyze threats
- Respond to attacks

Popular EDR Platforms

- CrowdStrike Falcon
- Microsoft Defender
- SentinelOne

What is XDR?

XDR stands for Extended Detection and Response.

XDR combines data from multiple security systems such as:

- Endpoints
- Email systems
- Firewalls
- Cloud platforms
- Networks

Advantages of XDR

- Better visibility
- Faster threat detection
- Centralized investigation
- Advanced automation
